

AIMScribe v3 — Integration & Test Guide for CMED DevOps

The endpoints to bind to, and how to prove the system is running · AIMS LAB, United International University · 9 October 2026

This is the operational companion to the **CMED Integration Specification**. That document says what the messages contain; this one says **where to send them, how to authenticate, and how to verify the system is up** — with commands your team can run today.

	What CMED's devops team needs to know
1	There are two endpoints on our cloud server for your backend, and one local connection on each clinic PC for your web page
2	One unauthenticated health endpoint proves the system is running. Poll it as often as you like
3	Authentication is a single header, X-CMED-Key, on the two backend endpoints only
4	The server is moving to DigitalOcean Bangalore — the lowest-latency region available to Dhaka. Section 2
5	CMED must send us two things before anything works: your page's exact web address, and your clinic codes. Section 11

Nothing CMED runs needs to change for the clinic PCs
The recorder listens on 127.0.0.1:5050 — the loopback interface of the clinic PC itself. No inbound firewall rule, no port forward and no network change is needed on CMED's side or the clinic's. Traffic never leaves the machine. Section 8.

THE ENDPOINTS

Exact addresses, method by method

Copy these. Everything else in this document explains them

0.1 The gateway

One hostname serves everything on the server side. AIMS LAB issues it with the key; it differs between staging and production, so hold it in configuration.

BASE_URL = https://<aims-host> <-- AIMS LAB sends this with the key

Transport HTTPS only, TLS 1.2 or 1.3. Port 443. HTTP redirects to HTTPS

Encoding JSON, UTF-8

Auth X-CMED-Key: <key> on the two clinical endpoints only

0.2 The four addresses CMED calls

#	Method	Full address	Auth	Body	Success
1	GET	https://<aims-host>/health	none	none	200
2	GET	https://<aims-host>/	none	none	200
3	POST	https://<aims-host>/api/v2/clinical/patient-information	X-CMED-Key	JSON ≤1 MB	202
4	POST	https://<aims-host>/api/v2/clinical/prescription	X-CMED-Key	JSON ≤1 MB	202

	Which API it is	Sent when
3	API 2 — patient information	the doctor opens a patient
4	API 3, Channel B — the prescription	the doctor presses Build Prescription

There is no GET for clinical data and there is no DELETE. CMED writes to us and never reads back: nothing clinical and no audio is ever served to CMED. Both clinical calls are POST, and a repeat of the same POST is safe — it returns 202 ALREADY_RECEIVED and stores one row.

0.3 The local address on each clinic PC

Protocol	Full address	Auth	Who
WebSocket	ws://127.0.0.1:5050/ws	page address allow-list	CMED's page, in the doctor's browser

Not HTTP and not a REST call — one WebSocket held open for the whole clinic session, carrying short JSON messages each way. **ws://** and not **wss://** because it never leaves the machine; loopback needs no certificate and a browser permits it.

Message CMED sends	Which API	Sent when
{"command": "start", "trigger": {...}}	API 1	the doctor opens a patient
{"command": "prescription_built", ...}	API 3, Channel A	the doctor presses Build Prescription
{"command": "doctors"}	—	on page load — returns this PC's clinic code
{"command": "status"}	—	optional; the recorder also pushes status unprompted

All six, on one line each

```
GET https://<aims-host>/health
GET https://<aims-host>/
POST https://<aims-host>/api/v2/clinical/patient-information
POST https://<aims-host>/api/v2/clinical/prescription
WS ws://127.0.0.1:5050/ws → command "start"
WS ws://127.0.0.1:5050/ws → command "prescription_built"
```

Everything else under /api/v2/ is closed to CMED's key — those endpoints belong to the recorders and use client certificates. There is nothing there CMED needs.

The server

Where it runs, how fast it answers, and what it is built on

1 · Addresses

	Address	Auth	Who calls it
Health	GET /health	none	CMED devops, monitoring, load balancers
Front page	GET /	none	a human, to confirm which system answered
API 2	POST /api/v2/clinical/patient-information	X-CMED-Key	CMED's backend
API 3, Channel B	POST /api/v2/clinical/prescription	X-CMED-Key	CMED's backend
API 1 & API 3, Channel A	ws://127.0.0.1:5050/ws	origin allow-list	CMED's web page, on the clinic PC
Everything else under /api/v2/	—	device certificates	the recorders only. Not CMED

CMED only ever calls three things: the health endpoint, and the two clinical endpoints. The rest of the /api/v2/ surface belongs to the recorders and is closed to CMED's key.

The hostname

AIMS LAB will send the final hostname with the key. Please treat it as configuration, not something compiled in — it will differ between your staging and production environments, and the staging host may be rebuilt. Throughout this document it is written as <aims-host>.

2 · Where the server runs, and how fast it answers

The system is being deployed to **DigitalOcean, Bangalore region (BLR1)**. Of all regions available to us, Bangalore is the closest to Dhaka and gives the lowest round trip.

Region	Distance from Dhaka	Typical round trip	Verdict
Bangalore — BLR1	~2,400 km	45–70 ms	Chosen
Singapore — SGP1	~3,000 km	60–95 ms	second choice
Frankfurt — FRA1	~7,400 km	130–170 ms	too far
New York — NYC3	~12,900 km	230–280 ms	too far

Latency does not affect recording. The microphone is driven by the program on the clinic PC, over loopback, so it starts in under a millisecond regardless of where our server is or whether it is reachable at all. The round trip to Bangalore only affects how quickly your two backend messages are acknowledged — and those are queued on your side anyway, so a slow reply is never visible to a doctor.

2.1 The machine

	Specification	Note
Provider and region	DigitalOcean, Bangalore (BLR1)	lowest ping to Dhaka

Droplet	Basic / Regular — s-8vcpu-16gb	the smallest standard droplet that meets every requirement
Processors	8 vCPU	as specified
Memory	16 GB	specification asked 8 GB; this tier includes 16
Disk	320 GB SSD, included	specification asked 200 GB+; no add-on volume needed
Outbound transfer	6 TB/month included	so uploads to Cloudflare R2 cost nothing
Operating system	Ubuntu Server 24.04 LTS	supported to 2029
Price	\$96/month, \$1,152/year	verify on DigitalOcean's pricing page

Why this droplet: 8 vCPU on DigitalOcean's Basic line is paired with 16 GB and 320 GB as a fixed bundle, so asking for 8 cores and 200 GB lands here and the extra memory and disk come at no additional cost. The 6 TB of included transfer matters more than it looks: the server pushes roughly 520 GB a month up to Cloudflare R2, and on this provider that is free.

2.2 Storage — the 320 GB disk is not where audio lives

The droplet's 320 GB is **working space**. It is not the audio archive, and it is nowhere near enough to be one. Every recording is copied to **Cloudflare R2 object storage** and then removed from the server.

	Server disk — 320 GB	Cloudflare R2 — 3 TB
Holds	Operating system, the eight programs, both databases, logs, and recordings in transit	Every recording, permanently , as encrypted FLAC
How long	Hours. A file is removed once the cloud copy is verified	Permanently . This is the research dataset
Capacity	about a day and a half of recording as a buffer	41,600 consultations — the 18,000-patient study uses 1.3 TB
Cost	included in the \$96 droplet	\$45/month, \$540/year
If it fills	archiving stops; clinics keep recording locally	buy more — \$15 per TB per month, no limit

Total cost of the cloud deployment	Monthly	Yearly
Droplet — 8 vCPU / 16 GB / 320 GB, Bangalore	\$96	\$1,152
Droplet backups	\$19	\$230
Cloudflare R2 — 3 TB of audio	\$45	\$540
Transfer to R2 (519 GB/mo, inside the included 6 TB)	\$0	\$0
Total	\$160	\$1,922

R2 charges nothing to read data back, which is why it was chosen: a research dataset gets downloaded many times, and on Amazon S3 each full read of 3 TB would cost about \$250.

2.3 The cloud copy keeps the same folder structure as the local archive

Object storage has no real folders — a key is just a long name — but a key may contain /, and every tool from Cloudflare's own browser to rclone displays those as folders. **So the layout is entirely ours to choose, and it already mirrors the local archive:**

ON THE SERVER

```
archive/AALO_DHOLPUR/DR0042/2026-10-05/<stem>/<stem>.wav
                                <stem>.json
                                <stem>.manifest.json
```

IN CLOUDFLARE R2

```
copies/AALO_DHOLPUR/DR0042/2026-10-05/<stem>.v1.flac.enc
copies/AALO_DHOLPUR/DR0042/2026-10-05/<stem>.v1.json.enc
```

where <stem> = P0012345_DR0042_AAL0_DH0LPUR_101432_102755_20261005

Same	Different	Why
clinic / doctor / date, in that order	a copies/ prefix	lets other things share the bucket later
the file name, character for character	.flac not .wav	lossless, 40% smaller — verified by decoding and comparing
	.enc on the end	encrypted before it leaves UIU — see below
	.v1 version number	a corrected copy never overwrites the first
	no per-consultation sub-folder	the stem is already unique, so the extra level earns nothing in a bucket

So you can browse R2 exactly as you browse the server — open a clinic, then a doctor, then a day, and the consultations are there. A restore can walk one clinic or one date without consulting any database, which is the reason the structure was chosen.

It is also ours to change. One function builds these keys (copy_object_key), so if a different arrangement suits the research better — year and month above clinic, say, or the patient above the date — it is a small change, not a migration. Worth settling before 18,000 recordings are in place rather than after.

One thing Cloudflare cannot do

Every file is encrypted on our server before it is uploaded (AES-GCM), and the key never leaves UIU. The folder names and file names are visible to Cloudflare; **the audio and the clinical record are not**. Cloudflare stores bytes it cannot read, and so could any future provider — which is what makes putting patient recordings on commodity storage acceptable at all.

Proving it is running

The health endpoint, and the exact commands to test with

3 · The health endpoint

This is the endpoint to monitor. No key, no body, no side effects. It checks the database and the queue on every call.

```
curl -s https://<aims-host>/health
```

A healthy system answers **HTTP 200** with:

```
{
  "status": "healthy",
  "database": "connected",
  "redis": "connected",
  "minio": "connected",
  "system": "AIMScribe v3",
  "srs": "3.x",
  "version": "3.x.x",
  "mode": "FastAPI Async"
}
```

Field	Watch for	What it means
status	healthy	Everything is up. Anything else is degraded
database	connected	If disconnected, nothing can be stored. Alert on this
redis	connected	The queue. If disconnected, archiving stalls
system	AIMScribe v3	Check this field. See the warning below
version	changes after a deploy	Useful to confirm a release landed

Please check the system field, not just the status code

There is an older **version 1** AIMScribe backend still running elsewhere, and it also answers on a `/health` path that looks almost identical. **Pointing at the wrong one costs an afternoon**, because every request succeeds and nothing is ever recorded. The system field is first in the response for exactly this reason — assert on it in your monitoring, not only on HTTP 200.

Monitoring guidance	Recommendation
How often	Every 30–60 seconds is ample
Alert when	<code>status != "healthy"</code> twice in a row, or no answer for 2 minutes
Timeout	5 seconds
Do not	Do not alert the clinics. A doctor cannot act on this, and recording continues regardless

4 · Which system answered — the front page

Opening `https://<aims-host>/` in a browser shows a one-screen page naming the system, its version, and where each party connects. It is the quickest way for a human to confirm they are pointed at AIMScribe v3 and not the version 1 backend. It reads no consultation data.

Binding to the two endpoints

Authentication, limits, and a test you can run before writing any code

5 · Authentication

	Detail
Header	X-CMED-Key: <key>
Applies to	The two clinical endpoints only. Not /health
Issued by	AIMS LAB, once, out of band. It is shown once and not recoverable — store it immediately
Scope	One key for CMED, covering all seven clinics
Rotation	Ask us; we issue the new key before revoking the old, so there is no window where both fail
Where it must live	Your server's secret store or environment. Never in page source, never in a repository, never in a mobile app

A wrong or missing key returns 401 INVALID_KEY. That is a hard stop, not something to retry — retrying a bad key just fills both our logs. Alert your team and contact us.

6 · Limits and behaviour

	Value	What happens at the limit
Body size	1 MB	413 TOO_LARGE. Usually means an image or file got into the JSON
Content type	application/json, UTF-8	400 MALFORMED_JSON
Success	HTTP 202	Stored before you are answered. A 202 means it is on disk
Duplicate send	HTTP 202 ALREADY_RECEIVED	Safe. One row, not two. Retry freely
Validation failure	HTTP 422 SCHEMA_INVALID	Stored in quarantine, not discarded. The reply lists the bad fields
Our server is down	500 / 503	Queue and retry with backoff. Never drop the message
Suggested client timeout	10 seconds	Then retry. The message is queued on your side, so this is never user-facing

There is no rate limit for CMED. Seven clinics at fourteen rooms produce roughly one message every two seconds at peak, which is far below anything we would need to throttle. Send as the consultations happen; just do it from a queue so a slow reply never reaches the doctor's screen.

7 · Test it now — four commands

These need no code and no page. Run them against the staging host as soon as you have the key.

7.1 Is it running?

```
curl -s https://<aims-host>/health | jq .
# expect: "status": "healthy" AND "system": "AIMScribe v3"
```

7.2 Is my key accepted? (send a patient)

```
curl -s -o /dev/null -w "%{http_code}\n" \
-X POST https://<aims-host>/api/v2/clinical/patient-information \
```

```
-H "X-CMED-Key: $CMED_KEY" \  
-H "Content-Type: application/json" \  
-d '{  
  "patient_id": "TEST0001",  
  "doctor_id": "DR0042",  
  "hospital_id": "AALO_DHOLPUR",  
  "start_time": "2026-10-09T10:14:32+06:00",  
  "date": "2026-10-09",  
  "demographics": { "name": "Test Patient", "sex": "female",  
                    "age_years": 34 },  
  "previous_visit": null  
}'  
# expect: 202
```

Send that exact command twice. Both times you should get **202** — the second is `ALREADY_RECEIVED`. That proves retries are safe, which is the single most useful thing to know before writing your queue.

7.3 Does validation work? (deliberately break it)

```
# previous_visit omitted on purpose, and sex is a code rather than a word
curl -s -X POST https://<aims-host>/api/v2/clinical/patient-information \
-H "X-CMED-Key: $CMED_KEY" -H "Content-Type: application/json" \
-d '{ "patient_id": "TEST0002", "doctor_id": "DR0042",
      "hospital_id": "AALO_DHOLPUR",
      "start_time": "2026-10-09T10:20:00+06:00", "date": "2026-10-09",
      "demographics": { "name": "Test", "sex": "F" } }' | jq .
# expect: 422 SCHEMA_INVALID, naming demographics.sex and previous_visit
```

The reply tells you which fields were wrong, by name. Build against that: it is faster than reading the specification, and the record is kept in quarantine rather than lost, so nothing is wasted by experimenting.

7.4 Send the prescription

```
curl -s -o /dev/null -w "%{http_code}\n" \
-X POST https://<aims-host>/api/v2/clinical/prescription \
-H "X-CMED-Key: $CMED_KEY" -H "Content-Type: application/json" \
-d '{
  "patient_id": "TEST0001",
  "doctor_id": "DR0042",
  "hospital_id": "AALO_DHOLPUR",
  "start_time": "2026-10-09T10:14:32+06:00", <-- SAME as API 2
  "date": "2026-10-09",
  "issued_at": "2026-10-09T10:26:55+06:00",
  "items": [ { "drug": "Napa 500 mg", "dose": "1 tablet",
               "frequency": "1+1+1", "duration": "5 days" } ],
  "diagnoses": ["Acute gastritis"],
  "investigations": []
}'
# expect: 202
```

The one thing these four commands cannot prove

All four can pass while the integration is still broken, because **a 202 means "stored", not "matched to a recording"**. The match is a separate step on our side that compares the five identifying fields from your message against the five the recorder reported. If start_time differs by one second, or loses its +06:00, every reply above is still green and **the recording is deleted twenty-four hours later**.

So the real end-to-end test needs a recorder running — section 9. Please do not treat four green curls as integration complete.

The clinic PC, and going live

The local channel, the end-to-end test, and what each side owes the other

8 · The clinic PC — nothing for devops to open

	Detail
What runs there	AIMScribe.exe, one per consulting room, installed by AIMS LAB
Listens on	127.0.0.1:5050 — loopback only
Reachable from	That PC alone. Not the LAN, not the internet, by design
Firewall changes needed	None , on CMED's side or the clinic's
Authentication	No key. The recorder checks your page's web address instead — see section 11
If it is not running	Your page's connection simply fails. Show "AIMScribe is not running on this PC"

Why no key here: a secret placed in a web page is readable by anyone who opens the page, so it would protect nothing. Instead the recorder accepts connections only from web addresses AIMS LAB has registered, only to loopback, and only from the same machine. **That is why your exact page address is a blocking prerequisite** rather than a detail.

9 · The end-to-end test that actually proves it works

	Step	Pass condition
1	AIMS LAB installs the recorder on one test PC and enrolls it	/health is healthy; the recorder's tray icon is running
2	Open your page on that PC and connect to ws://127.0.0.1:5050/ws	Connection opens. If refused, your address is not registered
3	Send {"command": "doctors"}	Returns the clinic code. Use it — do not hardcode one
4	Open a test patient: API 1 and API 2 together	RECORDING_STARTED or RECORDING_PROVISIONAL, and 202
5	Talk into the microphone for two minutes	The level field in the status messages moves above 0
6	Build a prescription: API 3 on both channels	GATE_ARMED, and 202
7	Open a second test patient	First recording closes, second starts, no gap
8	AIMS LAB confirms the recording was matched and archived	This is the only step that proves the integration. Ask us to confirm it explicitly

Step 8 is the one to insist on. Steps 1 to 7 can all pass while the five fields silently fail to match. Ask us to confirm, for your test patient, that a recording exists, was confirmed, and was archived — we can see that on the dashboard in a few seconds, and it is the only green light that means anything.

10 · Our test tooling, which you are welcome to use

	What it does	Command
1	Checks a message against every rule, with no server and no network	<code>python tools/channel_b_test.py --local</code>
2	Runs every case that matters against a live server — a good message, the same twice, a broken one, a changed prescription, a wrong key, an oversized body — and checks each reply	<code>python tools/channel_b_test.py --server https://<aims-host> --key <key></code>
3	Shows every message and reply side by side in a browser	open /protocol-test on the staging site

Please run number 2 and send us the output before the first clinic opens. It takes under a minute and it has caught every integration problem we have seen so far. AIMS LAB will send the script with the key.

11 · What each side must hand over

AIMS LAB gives CMED	When
The staging and production hostnames	with the key
The X-CMED-Key — shown once	on request, out of band
The seven clinic codes	with the key
The integration test script	with the key
Confirmation that a test recording matched and archived	on request, during step 8
CMED gives AIMS LAB — both are blocking	Why it blocks
1. The exact web address of your page, including staging and preview — for example <code>https://ehr.aaloclinic.com</code>	Without it the page cannot open the local connection at all. Nothing works. Note <code>www.</code> and non- <code>www</code> are different addresses, as are <code>http</code> and <code>https</code> and any non-default port
2. Your own clinic codes, so we can map them to ours	Without the mapping, a visit cannot be tied to the clinic it happened in. The record is stored, you receive a normal 202, and the recording is erased after 24 hours. The failure is completely silent from CMED's side, which is why it matters more than it sounds
A named technical contact	So a question in either direction does not wait a week
A staging environment we can send to	So we can test our side against yours

12 · Go-live checklist

	Item	Owner
■	Production hostname and key issued and stored in CMED's secret store	both
■	CMED's page addresses registered (production and staging)	CMED → AIMS LAB
■	Clinic code mapping agreed	CMED → AIMS LAB
■	/health in CMED's monitoring, asserting on system and status	CMED
■	Both endpoints sent from a durable queue with retry , not inline in the request that serves the doctor	CMED
■	The five fields built once per consultation and reused by all four messages	CMED
■	Clinic code read from the doctor's command, not hardcoded	CMED
■	Microphone level shown on the page	CMED
■	channel_b_test.py run and output sent to AIMS LAB	CMED
■	End-to-end test done, and step 8 confirmed by AIMS LAB	both
■	Server on DigitalOcean BLR1, backups enabled, R2 bucket live	AIMS LAB
■	Alerting on unconfirmed recordings	AIMS LAB

13 · One page for the CMED devops engineer

Question	Answer
How do I know it is up?	GET /health — no key. Expect status: healthy
What else must I assert on?	system == "AIMScribe v3" — there is an older v1 backend with a similar health path
How many endpoints do I call?	Three. Health, patient-information, prescription
How do I authenticate?	X-CMED-Key header, on the two clinical endpoints only
Where does the key live?	Your server's secret store. Never in a browser
Body limit?	1 MB. JSON, UTF-8
What is success?	HTTP 202 — stored before you are answered
Is retrying safe?	Yes. Same body twice gives one row and ALREADY_RECEIVED
What about 422?	Nothing is lost. Quarantined, and the reply names the bad fields
What about 500 / 503?	Queue and retry with backoff. Never drop
Rate limit?	None for CMED. Peak is about one message every two seconds
Do I open a firewall port for the clinic PCs?	No. The recorder is loopback-only
Where is the server?	DigitalOcean Bangalore (BLR1) , 8 vCPU / 16 GB / 320 GB, ~45–70 ms from Dhaka
Where does the audio live?	Cloudflare R2, 3 TB , as encrypted FLAC. Not on the 320 GB server disk , which is working space
Same folders as the server?	Yes — copies/clinic/doctor/date/
Does latency affect recording?	No. The microphone is driven locally
Does a 202 mean it worked?	No. It means stored, not matched. See section 9 step 8
What must I send AIMS LAB?	Your page's exact addresses, and your clinic codes. Both blocking

Prepared by AIMS LAB, United International University. Companion to the CMED Integration Specification, which carries the message-level detail. Every endpoint, limit and reply code here is taken from the running code. Prices and latency figures are indicative and should be confirmed before purchase.